

Technology Stack

Date	30 September 2026
Team ID	6ab3ecd3a238dc999eb3fcc2
Project Name	ComicCraft AI
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side <i>(e.g., React, Angular, HTML/CSS)</i>	HTML, CSS, Jinja2	Simple UI for collecting prompts and rendering comic previews with reusable templates.

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	Python, FastAPI, Uvicorn	Fast server-side routing, form/JSON processing and API orchestration.
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	No database in current version	Current workflow generates and exports comics without user-account or persistent database storage.
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	Local deployment	Project documentation specifies local FastAPI/Uvicorn deployment; cloud hosting is future work.
5	Version Control & CI/CD	Git	Version control and collaboration for source code and project changes.

6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	Google Gemini API + Hugging Face Diffusers / Stable Diffusion + FPDF	Gemini Flash creates structured 5-panel outlines; Gemini Pro writes narration/dialogue; Stable Diffusion creates panel illustrations; FPDF exports the final comic PDF.